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Pusher Syndrome – a frequent but little-known disturbance of body orientation perception

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■ **Abstract** Disturbances of body orientation perception after brain lesions may specifically relate to only one dimension of space. Stroke patients with “pusher syndrome” suffer from a severe misperception of their body’s orientation in the coronal (roll) plane. They experience their body as oriented ‘upright’ when it is in fact markedly tilted to one side. The patients use the unaffected arm or leg to actively push away from the unparalyzed side and resist any attempt to passively correct their tilted body posture. Although pusher patients are unable to correctly determine when their own body is oriented in an upright, vertical position, they seem to have no significant difficulty in determining the orientation of the surrounding visual world in relation to their own body. Pusher syndrome is a distinctive clinical disorder occurring characteristically after unilateral left or right brain lesions in the

posterior thalamus and –less frequently– in the insula and post-central gyrus. These structures thus seem to constitute crucial neural substrates controlling human (upright) body orientation in the coronal (roll) plane. A further disturbance of body orientation that predominantly affects a single dimension of space, namely the transverse (yaw) plane, is observed in stroke patients with spatial neglect. Apparently, our brain has evolved separate neural subsystems for perceiving and controlling body orientation in different dimensions of space.

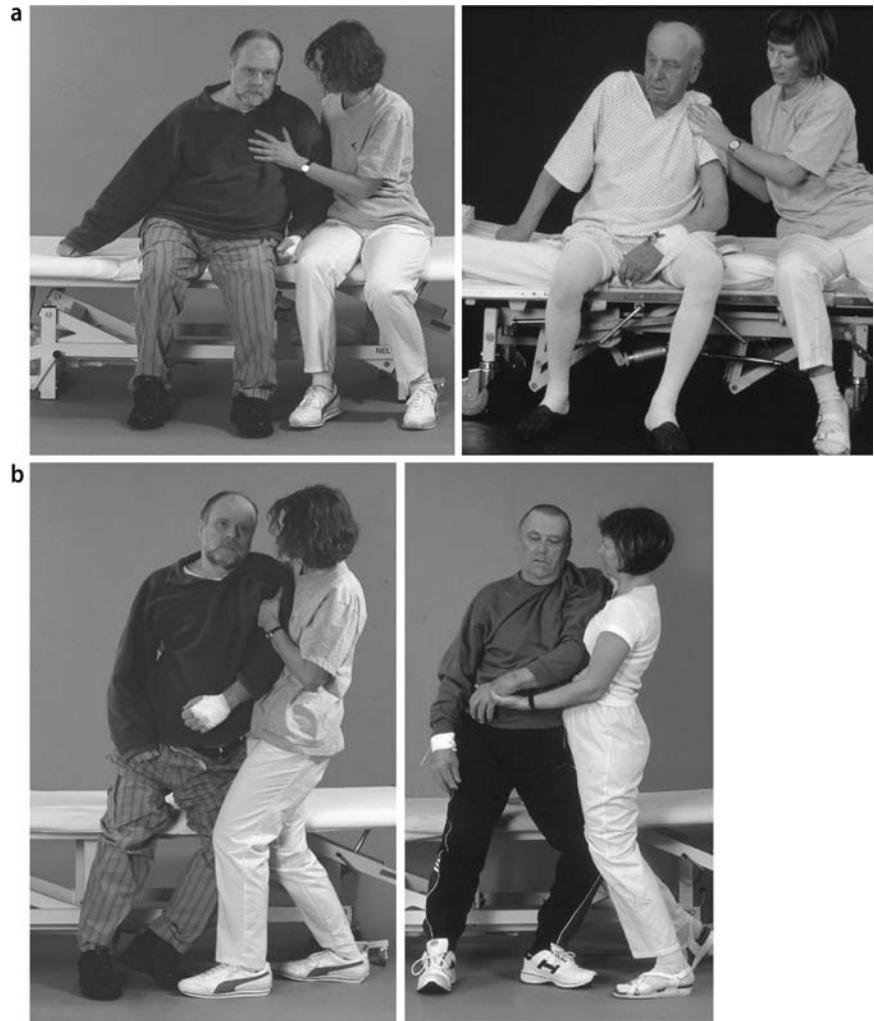
■ **Key words** pusher syndrome · vestibular · spatial orientation · spatial neglect · thalamus · insula · somatosensory processing · camptocormia · human

Introduction

Posture and balance impairments frequently occur in stroke patients and are due to different mechanisms. Deficits can be caused by pareses, sensory loss, or, e.g., by disturbed sensory integration or postural control [7, 20, 22, 27, 50, 62, 65]. One distinctive disorder of

postural control – termed the “pusher syndrome” [19] – has just recently come to the attention of clinicians and researchers. These patients use the non-parietic arm or leg to actively push away from their unparalyzed side (Fig. 1). If not prevented from doing so, they push themselves into such an unstable, laterally-tilted position that they fall toward the contralateral side. Characteristically these patients use the

Fig. 1 Stroke patients with pusher syndrome (here after right brain damage). The characteristic features of the disorder are that when sitting (a) or standing (b), these patients spread the non-paretic extremities from the body to actively push away from the non-paretic side (contraversive pushing). Moreover, they use these extremities to forcefully resist any attempt to passively correct their tilted body posture. (courtesy of Doris Broetz, Physiotherapy, University of Tuebingen)



non-paretic extremities to actively resist any attempt to passively correct their tilted body posture towards the earth-vertical upright orientation [19, 32].

Systematic investigations of large samples of acute stroke patients with hemiparesis found pusher syndrome in about 10% of the cases [51, 56]. More recent studies reported even higher percentages [18, 40], apparently due to less strict criteria for diagnosing the disorder. While the typical etiology of pusher syndrome is stroke, the disorder also occurs after other kinds of brain damage such as trauma or brain tumors [58].

Given the relative frequency of the disorder, why is it so little-known to clinicians compared with other posture and balance impairments? One reason might be the severe hemiparesis in these patients. When admitted to hospital, the contralesional arm of pusher patients is typically weak and the contralesional leg is so severely paretic that it cannot be moved against gravity [18, 28, 34, 36, 38]. When such patients are admitted to hospital, neurologists typically examine

them in the supine position. However, to detect a pusher syndrome, patients must be brought into an upright (sitting or standing) position (Fig. 1). Thus, quite often physiotherapists and nurses are first to detect the disorder, for instance when they transfer the patients from the bed to a wheelchair or while they treat them in a sitting position.

Bringing pusher patients into a sitting position in the acute stage of stroke infarct is a difficult maneuver that requires the assistance of at least one additional person (Fig. 1). Characteristically the patients do not collaborate, but instead act to counter the examiner's efforts.

New wine in old bottles?

Is the pusher syndrome a distinct neurological disorder or is it simply a new name for the postural control problems of stroke patients, known as e.g. the 'listing phenomenon', 'lateropulsion', or 'thalamic

astasia'? There are a number of differences distinguishing patients with pusher syndrome from patients with other neurological disturbances of balance and postural control, of which active pushing to one side and the resistance against passive correction are the most prominent.

■ Listing phenomenon

Because of the hemiparesis, any stroke patients may occasionally lose their lateral balance and fall toward the paralyzed side [4, 8, 13, 20, 55]. They may begin to list from an upright (sitting) position toward the hemiplegic side when assistance is withdrawn. This occasional loss of equilibrium in hemiparetic patients has been termed the 'listing phenomenon' [13], and was described as follows: "(The) patient is first assisted into a symmetrical sitting posture away from the back of the chair. When the assistance is withdrawn, the patient's trunk *begins to list toward the affected side, as if drawn by a magnet ...*" (p. 60 in ref. [13]). In contrast to patients with contraversive pushing, these patients recognize their loss of equilibrium, but – especially early after stroke onset – are unable to support themselves. Once they begin to cope with this situation, they usually *cling onto something with their non-paretic hand to prevent listing* [13]. These patients thus use their non-paretic hand to keep their bodies in an upright position, but not to actively push themselves into an unstable lateral tilt position, as is the case with pusher patients.

■ Lateropulsion

Active pushing away with non-paretic extremities also distinguishes pusher patients from patients with 'lateropulsion', a phenomenon commonly observed in Wallenberg's syndrome [6, 21]. Dieterich and Brandt [21] investigated 36 such patients with acute unilateral medullary brainstem infarctions and found that they had lateropulsion (defined as a tendency to fall sideways) with an *ipsiversive* deviation of the center of gravity (determined by means of posturography), i.e. with a deviation *towards* the side of the brain lesion. By contrast, patients with pusher syndrome tilt their body *contraversively* (towards the side opposite to the brain lesion), i.e., they use their unaffected arm to push towards the side of the hemiparesis [19, 32]. Thus, patients with lateropulsion and patients with pushing behavior tend to fall to opposite sides (*ipsiversively* in the case of lateropulsion and *contraversively* in the case of pusher syndrome). Moreover, patients with brainstem lesions and lateropulsion do not use their non-paretic extremities to actively push

away from and to resist passive correction towards the earth-vertical upright orientation.

■ Thalamic astasia

Does 'contraversive pushing' describe the same behavioral disorder that Masdeu and Gorelick [43] called 'thalamic astasia'? In their study, the authors reported 15 patients with unilateral, predominantly posterolateral thalamic lesions, who were unable to stand unsupported. Eight of the patients "could not even sit up by themselves and had marked truncal instability, falling backward or to the affected side from a sitting position when left without support. Typically, when [lying in bed and] asked to sit up, rather than using the axial muscles, these patients would grasp the side rail of the bed with the unaffected hand or with both hands to pull themselves up" (p. 597 in ref. [43]). Unfortunately, the authors did not describe the patients' behavior when sitting and standing. Thus, we do not know whether or not their patients showed the same pathological behavior that Davies [19] had defined 3 years before as 'pusher syndrome', i.e., whether they used their unaffected arm and/or leg to actively push away from the unparalyzed side and whether they resisted attempts to passively correct tilted body posture.

However, one observation suggests that Masdeu and Gorelick [43] did indeed observe a distinct disorder from what Davies [19] had called the pusher syndrome. When performing their study, the authors screened for and included only patients who had very mild or no motor weakness [J. Masdeu, personal communication]. Since unselected samples of pusher patients typically suffer from severe or complete paresis of their contralesional extremities [18, 28, 34, 36, 38], it might well be that Masdeu and Gorelick's study only included patients who – although they had lesions very similar to those of pusher patients (see below) – differed systematically from patients with pusher syndrome. To clarify this issue, it would be interesting to study whether the different symptoms described by Davies [19] and later by Masdeu and Gorelick [43] typically co-occur in acute stroke patients (especially in those suffering from posterior thalamic lesions) or whether they dissociate. In the latter case, one would expect that differences in lesion location exist within the thalamus.

The correlation between pushing behavior, aphasia and spatial neglect

The observation that pushing behavior after lesions affecting the right hemisphere is frequently associ-

tated with spatial neglect led to the hypothesis that pushing behavior might be caused by spatial neglect [18, 41, 57] and might thus be a right hemisphere syndrome. However, already Davies [19] had observed that pusher syndrome also occurs with left hemisphere lesions, where it is not associated with spatial neglect, but rather with aphasia. Subsequent studies confirmed Davies' observations and found that hemispatial neglect is *not* the cause of contraversive pushing. Although contraversive pushing is highly correlated with spatial neglect in patients with right hemisphere lesions (Table 1), spatial neglect is *not* observed in about 20% of right brain-damaged patients with contraversive pushing and never seen in left brain-damaged patients with contraversive pushing [34]. Rather, all patients with pusher syndrome due to left-sided brain lesions also suffered from aphasia [34]. The strong correlations between pushing behavior and neglect after right hemisphere lesion and between pushing behavior and aphasia after left hemisphere lesion have been replicated in various samples of stroke patients [12, 18, 28, 36, 38, 40, 58, 60].

The neural basis of pusher syndrome

The strong correlation between contraversive pushing, aphasia, and spatial neglect is due to close anatomical relationships at the cortical and subcortical levels. Pedersen et al. [51] found contraversive pushing as often after left and as after right hemisphere damage. Other groups found that it occurred slightly more often after right than after left brain damage [18, 34, 38] with around 60% of patients with the pusher syndrome presenting a right hemispheric stroke versus 40% with left hemispheric strokes.

To identify the neural structures within the left and the right hemispheres that are associated with the pusher syndrome, Karnath et al. [34] investigated a sample of 46 consecutive patients suffering from cortical and/or subcortical lesions. The analysis re-

vealed that in left as well as in right brain-damaged patients with the pusher syndrome, it was typically the posterolateral thalamus that was damaged. The finding suggested that the posterior thalamus plays a fundamental role in controlling upright body posture. This conclusion was corroborated by a recent study of thalamic stroke patients that were prospectively investigated over a three-year period [38]. While strokes in the left as well as in the right posterior thalamus provoked pusher syndrome, lesion of the anterior thalamus typically did not cause the disorder.

Clinical observations suggest that the posterior thalamus is not the only neural substrate relevant for our perception of upright body orientation. Although less frequently, pusher syndrome is also observed in patients with brain lesions sparing the thalamus. Johannsen et al. [28] thus evaluated 45 acute patients with and without contraversive pushing following left- or right-sided lesions without additional involvement of the thalamus. In both hemispheres, the area of lesion overlap associated with contraversive pushing was typically centered on the insular cortex and parts of the postcentral gyrus. However, in direct contrast to matched, brain-damaged controls, the authors found no area of lesion overlap that was exclusively associated with pusher patients. This indicates that the structures involved in controlling our upright body orientation at the cortical level are in close proximity to those areas that induce aphasia in the left hemisphere and spatial neglect in the right hemisphere.

In conclusion, it seems as if the posterior thalamus, as well as parts of the insula and postcentral gyrus, are the neural structures that process the afferent sensory signals mediating graviceptive information about upright body orientation in humans. In fact, anatomical findings in the macaque monkey provide evidence for a close anatomical relationship between these neural structures. Thalamocortical axons arising in the ventral posterolateral and posteromedial nuclei project to the primary somatosensory cortex in the postcentral gyrus (Brodmann areas 3a, 3b, 1, and 2), to the secondary somatosensory cortex in the parietal operculum, and to the insula [23, 31].

Table 1 Frequency of additional symptoms occurring with pushing behavior after right-sided (R) and after left-sided (L) brain lesions (from ref. [34]).

		Pusher syndrome	
		R	L
Paresis of contralesional side	% present	100 %	100 %
Arm (<i>Strength*</i>)	Median	0	0
Leg (<i>Strength*</i>)	Median	2.5	3
Aphasia	% present	7 %	100 %
Spatial neglect	% present	80 %	0 %

* The degree of paresis of the upper and lower limbs was scored with the usual clinical ordinal scale, where '0' stands for no trace of movement and '5' for normal movement.

The pathogenesis of contraversive pushing

The pusher syndrome is a transient phenomenon that recovers within several weeks in the majority of stroke patients [12, 18]. Pushing behavior had resolved in 79% of affected patients within three months after acute stroke on one study [18]; patients of another study had nearly completely recovered by 6 month after stroke onset [36]. This demonstrates two points:

(i) that the disturbed mechanism underlying pusher syndrome is well-compensated for by our brain and (ii) that studies aiming to investigate these mechanisms need to be carried out very early after stroke onset. Unfortunately, experiments in this acute phase of stroke are difficult to perform since patients in this stage are severely handicapped and usually cannot keep their attention during longer investigations. Therefore, only few studies have addressed the mechanisms underlying pusher syndrome so far.

■ Subjective visual vertical

An obvious hypothesis to explain the disorder is that it is caused by a disturbance of the visuo-vestibular system. One way to test this is to measure subjects' perception of the subjective visual vertical (SVV). The SVV provides a sensitive and direction-specific measurement of peripheral and central vestibular dysfunction at different levels from the vestibular organ, through the brainstem and thalamus, up to the cortical level [24, 10, 11, 64]. In complete darkness, a luminous rod is presented at eye level in front of the upright, seated subject. The rod is placed at a random angular offset in the coronal (roll) plane and the subject is asked to rotate the tilted rod so that it appears to be earth-vertical.

Surprisingly the first study investigating the perception of the SVV in pusher patients found *undisturbed* processing of visual and vestibular inputs [35]. The authors measured an average SVV orientation of -0.4° in their pusher patients, a value similar to that measured in control subjects. Correspondingly, they observed that patients with pusher syndrome were able to align their bodies to earth-vertical structures in the laboratory's visual surroundings perfectly well (Fig. 2b). Even when tilted in a position which they subjectively perceived as 'upright' (see below), these patients could correctly determine visual vertical [35]. Subsequent studies have found small tilts of the SVV in pusher patients in the order of 3.2° and 4.8° [29, 60], which is well within the range of 4.5° to 6° observed for left and for right brain-damaged patients *without* the pusher syndrome [11, 39, 61, 66].

Thus, pusher patients obviously do not show a SVV tilt which is characteristic of only pusher syndrome. Moreover, the slight angles observed in pusher patients can hardly account for their dramatic tilting behaviour observed when these patients push their own bodies to such extreme lateral tilt angles that they fall over to that side. A recent study using rotation and caloric stimulation supplemented these findings by showing that dysfunction of the semicircular canals was not relevant for the clinical manifestation of pushing behaviour [59]. Visual-vestibular

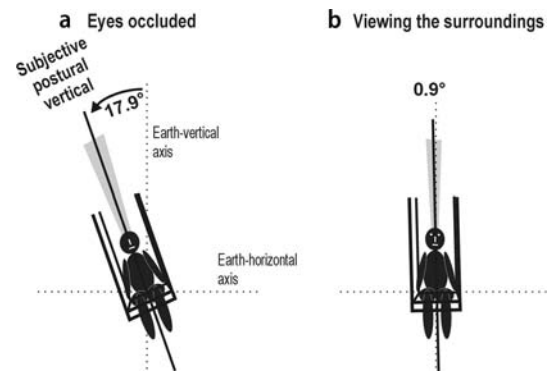


Fig. 2 Sitting on a lateral tilt chair, patients with pusher syndrome were required to indicate when they reached 'upright' body orientation. (a) With occluded eyes, the patients experienced their body as oriented 'upright' when they were actually tilted by nearly 20° towards the side of the brain lesion. (b) While viewing the laboratory's surroundings with many vertical structures (door, cupboard etc.), the same patients were able to align their longitudinal body axis to earth-vertical. (from ref. [35])

dysfunction and a disturbed perception of the visual world's orientation thus do not suffice to explain the disturbance.

■ Subjective postural vertical

While the SVV provides a sensitive measure of vestibular dysfunction, the task of adjusting one's subjective postural vertical (SPV) reflects the perceived upright orientation of the body. In darkness or with eyes occluded, subjects are seated upright on a lateral tilt chair [25, 42, 63]. They are immobilised by lateral stabilisation and are seated such that their feet do not touch the ground. After being rotated to a random starting offset in the frontal plane, subjects are asked to indicate when repositioning the chair brings their body in an earth-vertical upright position. Interestingly, neither cortical nor peripheral lesions of the vestibular system affect correct perception of the SPV [1, 5, 15, 25]. Despite concurrent vertigo, vection, and tilt of the SVV, patients with e.g. vestibular neuritis or patients who have undergone vestibular neurectomy, as well as healthy subjects receiving unilateral vestibular stimulation show undisturbed perception of body verticality (SPV) [5].

Karnath and co-workers [35] examined the SPV of acute stroke patients with pusher syndrome on a lateral tilt chair. In clear contrast to the findings in patients with vestibular disorders, they found a marked tilt of the SPV. On average, patients with pusher syndrome experienced their body as oriented 'upright' when they were actually tilted by nearly 20° to the side (Fig. 2a). This finding indicates that pusher syndrome is associated with a severe misperception of body orientation in relation to gravity.

The authors' *a priori* working hypothesis in that study was that pusher patients – when sitting on the edge of their bed – push their body towards a position in which they subjectively perceive themselves to be upright, thus expecting to find a tilt of the SPV towards the contralesional side. However, when investigating stroke patients as early as 13 days post-stroke, i.e. at a time when all these patients showed marked pushing behavior, they observed a marked tilt of the SPV towards the ipsilesional side. This counterintuitive finding raised the question of how an *ipsiversively* tilted perception of body orientation could lead to *contraversive* pushing activity. Two possible explanations are as follows:

Generally, a conflict between two reference systems is either resolved by suppressing one of them, or both, or by a compromise, e.g., by weighted summation. However, none of these seems to occur in the present situation. Under normal bedside conditions, the pusher patients align their body neither with the visual vertical, nor with their perceived postural vertical, or with an intermediate posture. Rather, they move their body into the opposite direction. By pushing their longitudinal body axis towards the contralesional side, the patients might be trying to actively compensate for the mismatch between visual vertical and the (tilted) orientation of body verticality. The clinical observation that patients with contraversive pushing diminish their pushing behavior when visual input is excluded (with eyes closed) supports this notion.

Alternatively, it is possible that the pushing behavior is a secondary response to the patients' unexpected experience that they lose lateral balance when trying to get up and sit upright in an upright room. The patients' subjectively perceived 'upright' orientation is tilted by nearly 20° towards the ipsilesional side [35]. Thus, when the patients try to get up and move their body to a (subjectively) 'upright' position, they become laterally unstable due to the fact that the center of mass is shifted too far to the ipsilesional side. Pushing the body to the opposite (contralesional) side might constitute a reaction to this experience.

Future studies are needed to investigate these as well as other possible interpretations.

A second graviceptive system in humans

Whatever the explanation for the unexpected *direction* of the SPV tilt towards the ipsi(not contra)lesional side may be, it has become clear that the pusher syndrome is associated with a severe misperception of body orientation in relation to gravity [35]. Moreover,

it has become obvious that a disturbance in perceiving the orientation of the visual world does not suffice to explain this disorder [29, 35]. Interestingly, patients with lesions of the vestibular system behave in exactly the opposite manner. They exhibit visual-vestibular dysfunction with a perceptual tilt of the visual vertical, but have no difficulty orienting their body to an earth-vertical, upright position [1, 5, 15, 25]. This double dissociation argues for a neural pathway in humans for sensing the orientation of gravity and controlling upright body posture, separate from the well-known visual-vestibular system for perceiving the orientation of the visual world.

What might be the sensory input for this second graviceptive system? The various somatosensory inputs from the skin and muscle tendons may constitute one component. Bisdorff et al. [5] even speculated that proprioception alone may be sufficient for a reasonable estimate of uprightness. However, proprioceptive input from the lower extremities appears to exert only an indirect influence on the perception and control of posture. For example, when astronauts are in space for the first time and are instructed to keep their body perpendicular to the floor of the space station (with restricted vision), the body leans forward by nearly 20° even though the astronauts perceive themselves to be perfectly perpendicular to the floor [16, 17]. The same forward inclination of nearly 20° was also observed under water when professional divers were instructed to adopt a vertical posture under water with their feet fixed to the ground [44]. These results show that somatosensory information, e.g., from ankle joints, is not enough to enable precise upright localization of body orientation with respect to the ground plane.

This conclusion is corroborated by findings in a patient with a complete loss of somatosensory input from one side of the body due to a right thalamic lesion. Despite his complete left-sided sensory loss, this patient did not have any difficulties adjusting the SPV veridical in a lateral tilt chair [35]. Comparable observations have been made in healthy subjects when somatosensory input was suppressed by cooling or local anaesthesia of the buttocks [2, 25]. Thus, it appears that somatosensory input plays a relatively minor role in our perception of body posture and that somatosensory loss does not necessarily result in pusher syndrome.

Mittelstaedt [46] proposed that the orientation of the visual world and of the head relative to the vertical is exclusively perceived through our (visual, vestibular, proprioceptive) sense organs in the head and the neck, while the trunk posture is perceived mainly through sense organs in the trunk. As possible candidates for truncal graviceptors in humans, he assumed the afferent input from the kidneys via



Fig. 3 Patient with right hemisphere stroke and pusher syndrome. In contrast to stroke patients without pushing behavior and patients with acute unilateral vestibular loss, the non-paretic, right leg of the pusher patient is constantly tilted ipsiversively (with respect to his trunk orientation). (from ref. [30])

the renal nerve and the information received through the inertia of a mass in the body. The blood in the large vessels or the mass of the abdominal viscera were thought to contribute to the latter afferent input transmitted via the phrenic or the vagus nerves. If this hypothesis holds, the areas in the posterior thalamus, insula and postcentral gyrus that cause pusher syndrome when lesioned [28, 34, 38] might represent those structures in which the afferent sensory signals from these postulated truncal sensors are processed. Functional imaging techniques in humans have indeed shown that stimulation of the vagus nerve influences neural activity in the thalamus, the insular cortex and the postcentral gyrus, among other brain regions [9, 14, 48].

However, posterior thalamus, insular and postcentral cortex are also involved in processing sensory input from our skin, muscle tendons, etc. Thus, the anatomical findings in pusher patients could also indicate that afferent somatosensory input from the traditional, well-known receptor systems provides the basis for the second graviceptive system in humans.

Disturbed spontaneous postural responses

Healthy individuals spontaneously adjust their posture in response to perturbations of balance. In pusher patients, whose perception of own body orientation is severely disturbed [35], one can expect these postural reactions in response to such pertur-

bations to be impaired. An initial study of 3 pusher patients did indeed find this to be the case. Pérennou et al. [54] investigated the maintenance of sitting posture under dynamic, unstable conditions on a freely moving seesaw (a so-called 'rocking platform'). Sitting without lateral stabilisation, subjects were responsible for both their own imbalance and active correction of the imbalance. In this self-perturbed and self-regulated dynamic task, the authors measured the orientation of the patient's head, shoulders, thoracolumbar spine, and pelvis and counted how often they lost their balance when sitting on the 'rocking platform' seesaw¹. In this experimental condition, the 3 patients with pusher syndrome had trouble maintaining an upright position and they fell off the seesaw onto the armrest and showed a pelvic tilt. However, postural instability while sitting on this seesaw is not unique for only patients with pusher syndrome. Using the same 'rocking platform' device, the investigators found that stroke patients with right and with left brain damage *without* pusher syndrome had comparable difficulties [52, 53].

Nevertheless, there also seem to be disturbances of spontaneous postural responses that indeed are specific for patients with pusher syndrome [30]. When investigating spontaneous postural responses of the non-paretic leg of pusher patients during passive sideways body tilts, this limb was found constantly tilted ipsiversively (with respect to trunk orientation) as compared with stroke patients without pushing behaviour and patients with acute unilateral vestibular loss (Fig. 3). Across the whole tilt range, the non-paretic leg of the pusher patients showed a constant ipsiversive tilt for an amount which was observed in the non-pusher subjects when they were tilted for about 15° into the ipsiversive direction [30]. Figure 4 illustrates the relationship between trunk orientation of the patients with pushing behaviour and the controls for a given leg orientation. For example, while sitting upright the pusher patients demonstrated a spontaneous orientation of the ipsilateral leg that was shown by the non-pusher subjects when tilted about 20° to the ipsiversive side (Fig. 4).

¹Pérennou and co-workers termed this measure the 'behavioural vertical'. Unfortunately, some recent studies misinterpreted this measure as reflecting the traditional 'subjective postural vertical (SPV)'. Obviously, the close terminology induced this confusion. However, in contrast to the 'rocking platform' devices in the SPV paradigm, blindfolded subjects are seated on a lateral tilt chair that is passively tilted by the experimenter while subjects are immobilised by lateral stabilisation and indicate verbally or by button press when the chair brings their immobilised body to an 'upright' position [15, 25, 42, 63]. Discrepancies in behaviour when measuring the 'behavioral vertical' and the 'subjective postural vertical (SPV)' are thus to be expected.

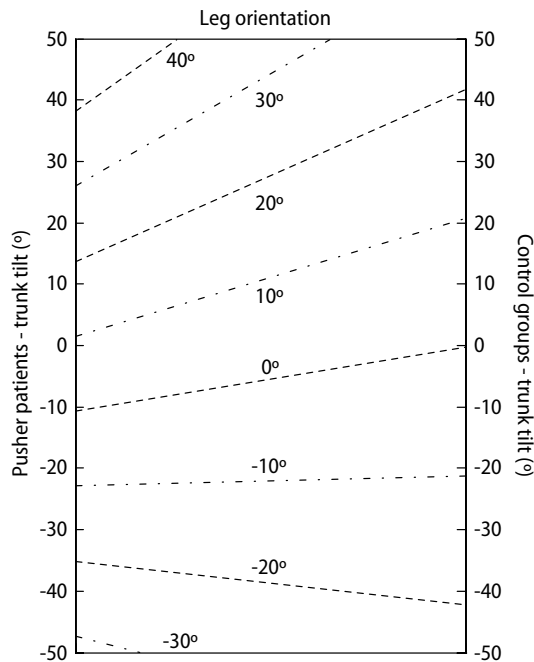


Fig. 4 Relation of leg-to-trunk orientation between the patients with pusher syndrome and the non-pusher control subjects investigated by Johannsen et al. [30]. The figure allows one to estimate the pusher patients' perceived subjective postural vertical from the appropriate postural adjustments of the non-paretic leg during passive trunk tilt. For example, while sitting upright the pusher patients demonstrated a spontaneous orientation of the ipsilateral leg that was shown by the non-pusher subjects when their trunk was tilted about 20° to the ipsiversive side. The pusher patients' leg-to-trunk orientation thus is indicative of an ipsiversive tilt of perceived body orientation of about 20°. (courtesy of Leif Johannsen, University of Birmingham)

The finding that patients with acute unilateral vestibular loss showed no such alterations of leg posture indicated that the disordered leg response of pusher patients is not the result of disturbed vestibular input. Johannsen and co-workers [30] thus concluded that in pusher patients a representation of body orientation might be disturbed that drives both conscious perception of body orientation and spontaneous postural adjustment of the non-paretic leg in the roll plane.

Specific subsystems for postural control in different dimensions of space

The experiments carried out so far have revealed that patients with pusher syndrome suffer from a severe misperception of body orientation in the coronal (roll) plane. They perceive their body to be oriented 'upright' when it is in fact markedly tilted to one side. The posterior thalamus, as well as parts of the insula and postcentral gyrus, seem to constitute crucial

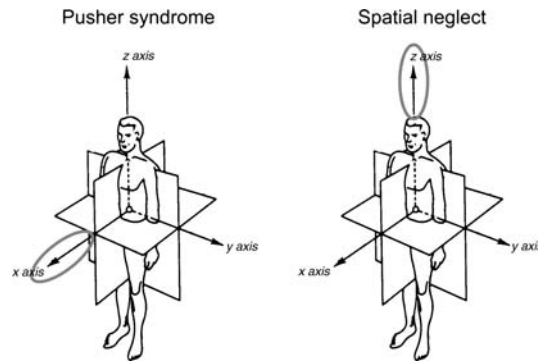


Fig. 5 Patients with pusher syndrome suffer from a severe misperception of body orientation in the coronal (roll) plane. They experience their body as oriented 'upright' when it is in fact markedly tilted to one side. (The red circle indicates the axis of pathological perception of body orientation.) In stroke patients with spatial neglect, a systematic disturbance of body orientation specifically affects the transverse (yaw) plane. These patients exhibit a spontaneous bias of eyes and head along the horizontal dimension of space leading to neglect of objects or persons on the left. The neural correlates of pusher syndrome and of spatial neglect thus seem to represent distinct networks for perceiving and controlling body orientation in different dimensions of space

neural substrates in humans for controlling (upright) body orientation in the coronal (roll) plane.

A further disturbance of body orientation which predominantly affects only one dimension of space, namely the transverse (yaw) plane (Fig. 5), is seen in stroke patients with spatial neglect. These patients exhibit a spontaneous bias in eye and head position along the horizontal dimension of space, which leads to a marked bias in active motor behavior towards the right and neglect of objects or persons on the left (for review ref. [33]). Damage to the right superior temporal cortex, insula, and temporo-parietal junction (TPJ) correlate with spatial neglect [26, 37, 47].

The neural correlates of pusher syndrome and of spatial neglect thus seem to represent distinct networks for perceiving and adjusting body position relative to external space. The two networks appear to control human posture in two different spatial dimensions, namely in the coronal (roll) plane and in the transverse (yaw) plane (Fig. 5). Thus, it would be interesting to ascertain whether circumscribed brain damage also can lead to specific disturbance of body orientation related to the third, sagittal (pitch) plane. Abnormal trunk posture with marked bending forward which improves when the patient is in supine position has indeed been reported [3, 49] and has been termed 'camptocormia'. However, the pathogenesis of such latter disturbance is still unclear.

In conclusion, it seems as if our brain has evolved separate, though anatomically partly overlapping, neural subsystems for perceiving and controlling body orientation in the different dimensions of space.

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